Dehydration Impairs Cognitive Performance: A Meta-analysis

MATTHEW T. WITTBRODT and MELINDA MILLARD-STAFFORD

Exercise Physiology Laboratory, School of Biological Sciences, Georgia Institute of Technology, Atlanta, GA

ABSTRACT

WITTBRODT, M. T., and M. MILLARD-STAFFORD. Dehydration Impairs Cognitive Performance: A Meta-analysis. *Med. Sci. Sports Exerc.*, Vol. 50, No. 11, pp. 2360–2368, 2018. Dehydration (DEH) is believed to impair cognitive performance but which domains are affected and at what magnitude of body mass loss (BML) remains unclear. **Purpose**: To conduct systematic literature review and meta-analysis to determine the effect size (ES) of DEH on cognitive performance and influence of experimental design factors (e.g., DEH > 2% BML). **Methods**: Thirty-three studies were identified, providing 280 ES estimates from 413 subjects with DEH ranging from 1% to 6% BML. Outcome variables (accuracy, reaction time), cognitive domains, and methods to induce DEH varied. Effect sizes were calculated using standardized mean differences and multivariate meta-analysis. **Results**: Impairment of cognitive performance (all domains/outcomes) with DEH was small but significant (ES = −0.21; 95% confidence interval [CI]: −0.31 to −0.11; $P < 0.0001$) with significant heterogeneity ($Q(279) = 696.0$, $P < 0.0001$; $I^2 = 37.6\%$). Tasks of executive function (ES = −0.24; 95% CI: −0.37 to −0.12), attention (ES = −0.52; 95% CI: −0.66 to −0.37), and motor coordination (ES = −0.40 to 95% CI: −0.63 to −0.17) were significantly impaired ($P \leq 0.01$) after DEH, and attention/motor coordination was different ($P < 0.001$) from reaction time specific tasks (ES = −0.10; 95% CI: −0.23 to 0.02). Body mass loss was associated with the ES for cognitive impairment ($P = 0.04$); consequently, impairment was greater ($P = 0.04$) for studies reporting >2% BML (ES = −0.28; 95% CI: −0.41 to −0.16) compared with ≤2%; (ES = −0.14; 95% CI: −0.27 to 0.00). **Conclusions**: Despite variability among studies, DEH impairs cognitive performance, particularly for tasks involving attention, executive function, and motor coordination when water deficits exceed 2% BML. **Key Words:** HYPOHYDRATION, EXECUTIVE FUNCTION, ACCURACY, REACTION TIME

Dehydration (DEH) has known adverse effects on the human body (1). It is well documented that physical performance tasks (aerobic exercise, muscular endurance, occupational tasks, sport-specific tasks) are impaired with DEH (1–4). In contrast, whether cognitive performance is also susceptible to DEH and at what threshold of body water loss is far less clearly defined (5). Cognitive performance is a measurable outcome (e.g., accuracy, reaction time) during tasks requiring decision making, problem solving, attention, judgment, memory or eye–hand coordination. Initial studies on DEH and cognitive performance suggested executive function and information processing were impaired after DEH of 2% body mass loss (BML) (6,7). However, these findings have not been uniformly supported in subsequent studies (8–10). No sole reason accounts for the equivocal findings within the literature; however, potential variables include differences across methods to elicit DEH (e.g., exercise, exercise-heat stress, fluid restriction, diuretics), the magnitude of DEH, and the specific cognitive task evaluated (11–13). Although narrative reviews highlight potential factors influencing the cognitive responses to DEH (12–15), a quantitative analysis that systematically examines the effect of these variables is absent from the literature.

It is clear that severe levels of DEH (e.g., >8% BML) elicit discernible cognitive impairments (16,17). Soldiers in adverse environments (e.g., desert heat with extended water restriction) have an impaired ability to navigate, successfully complete military operations, and, if DEH is severe enough, present with confusion and delirium (16,17). Soldiers undergoing 5% BML (solid and liquid) during a 72-h training exercise had impaired (by twofold to fourfold) vigilance, reaction time, attention, memory, and reasoning compared with their performance at rest (18). However, these field-based military studies inducing large magnitudes of DEH typically include other cofactors also known to alter cognitive performance, including sleep deprivation (19), hypoglycemia (20), and other physiological stressors (21).

Thus, there is no clear threshold established at what magnitude of DEH (e.g., ≥2% BML) cognitive impairments begin to occur. Because 2% BML elicits physical (e.g., aerobic) performance decrements (1) along with accompanying

Address for correspondence: Mindy Millard-Stafford, Ph.D., 555 14th St NW, School of Biological Sciences, Georgia Tech, Atlanta, GA 30332; E-mail: mindy.millardstafford@ap.gatech.edu.
Submitted for publication March 2018.
Accepted for publication May 2018.
Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal's Web site (www.acsm-msse.org).

0195-9131/18/5011-2360/0
MEDICINE & SCIENCE IN SPORTS & EXERCISE®
Copyright © 2018 by the American College of Sports Medicine
DOI: 10.1249/MSS.0000000000001682

APPLIED SCIENCES

Copyright © 2018 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

physiological compensation due to hypovolemia and increased plasma osmolality (5), some suggest that cognitive impairments also begin to arise in parallel. Experimental evidence indicates impairments occur at 2% with exacerbated decrements in cognitive functions at 4% BML (6,7), suggesting an association between body water deficits and impaired mental functioning. However, not all subsequent studies have supported this relationship, with a recent review (15) suggesting additional protective mechanisms for the brain to preserve cognitive functions until DEH reach a higher threshold (e.g., ≥3% BML).

Therefore, our purpose was to perform a systematic review of the literature and use a quantitative technique (i.e., meta-analysis) to determine the impact of DEH on performance of cognitive tasks. Our primary aim was to examine potential experimental design factors (e.g., method to elicit and magnitude of DEH, and cognitive test domain) that may influence the effect size (ES) estimate. We hypothesize that, like previous narrative reviews, DEH will induce a small but significant impairment in cognitive performance. Second, we hypothesize that the magnitude of DEH will be significantly associated with the degree of cognitive impairment, observable at a minimum threshold (>2% BML) similar to that of other physical performance measures.

METHODS

A systematic review was conducted on the research literature for the effects of DEH and cognitive performance. Cognitive performance was operationally defined as any measurable outcome resulting from completion of a cognitive function task (e.g., reaction time, accuracy). The literature search was completed as of September 13, 2017. Searches were conducted in the following databases: PubMed, Medline, Psych Info, SportDiscus, ISI Web of Science, SCOPUS, ProQuest Theses and Dissertations, which collectively returned 8306 results (6591 without duplicates). References from relevant review articles were also examined (11,13,15,22) for articles not uncovered previously. Search terms consisted of: (*hydration OR water loss OR weight loss OR hypovol* OR sweat loss) AND (cognition OR cognitive function OR cognitive performance OR executive function OR response time OR reaction time OR intelligence OR memory OR mood OR vigilance OR pattern recognition OR letter* OR processing) AND (adult* OR college student).

Inclusion criteria. Studies meeting the following inclusion criteria were considered for review: (i) the study was conducted on healthy (i.e., no clinical conditions) adults (≥18 yr), (ii) the study contained at least two timepoints (within or between groups) when cognitive testing was completed after DEH and under a control condition, (iii) changes in hydration status were reported with BML, and (iv) cognitive performance variables (e.g., accuracy, reaction time) were reported. We did not extract data related to mood because this psychological construct did not fit the operational definition of cognitive performance with accuracy or reaction time outcomes. Studies not on healthy adults and those inducing chronic DEH (beyond 72 h) were excluded from the analysis. Studies were not excluded, however, due to elements of the research design (e.g., subject familiarization protocols, randomization of trial order, or type of control trial used for comparison).

Selection of studies. A total of 6591 relevant publications were originally identified through the database searches.

Of those, 6512 were initially excluded based on title and/or review of the abstract Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA [PRISMA diagram, Fig. 1]). Therefore, the full text of 79 studies was reviewed for meeting the inclusion criteria. Of those 79 studies, 48 were excluded due to no control condition, weight loss induced by >3 d fluid restriction, no BML measure, and no behavioral measures of cognitive performance resulting from DEH. Our screening criteria resulted in a total of 33 articles to be included.

Data extraction. Studies included in the meta-analysis were independently coded by a minimum of two reviewers. Discrepancies in data entry were discussed and a consensus was reached. Means, standard deviations, sample size, and correlations (if available) for both DEH and control conditions for all cognitive tasks within the study were extracted. If a study included any treatment condition other than DEH and control, the data for those conditions were omitted. Each task was categorized into specific cognitive domains of: attention, executive function, memory (short-term, working, long-term), information processing, motor coordination, or reaction time-specific tests according to previously published criteria (23,24) and/or author description of the task. Cognitive outcome variables (e.g., reaction time, accuracy) were extracted for all ES. If the outcome variable of a given cognitive test was not explicitly described as reaction time or accuracy, it was categorized according to the attribute most closely aligning (e.g., errors as accuracy, speed as reaction time). Study quality scores (e.g., PEDro) were not calculated

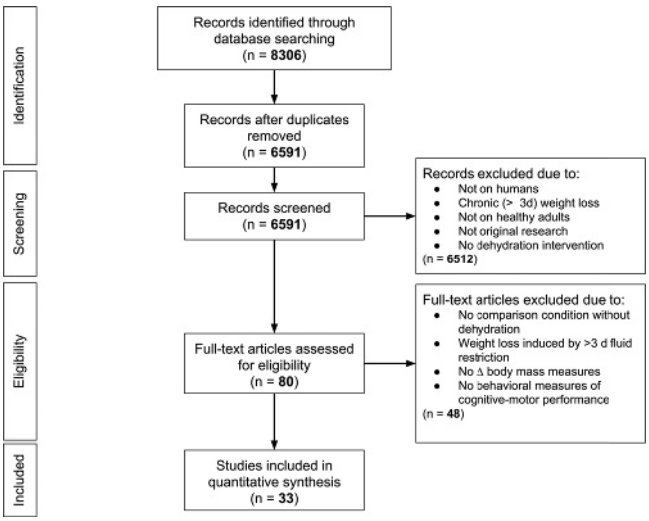

FIGURE 1—Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) diagram depicting the systematic review protocol in determining the inclusion of studies within the meta-analysis.

APPLIED SCIENCES

Copyright © 2018 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

because many of these studies omitted descriptions for specific design elements (e.g., blinding of subjects/therapists, group allocation) present in clinical trials for which these metrics were based on.

Meta-analysis. The extracted cognitive performance data were converted to a standard format by calculating the standardized mean change score or ES using the *metafor* package for R (v1.9-9, www.metafor-project.org). In studies where the correlational data were not reported, r was estimated from the median correlation taken from studies with (i) known DEH–control correlations (25) and studies reporting ES, means, and SD from which r could be calculated (26,27). The known correlations (n = 15; range, 0.01–0.92) had a median r of 0.62. For the ES estimate, Hedges g was used to minimize the inherent bias of Cohen d to overestimate the ES when standardized mean differences are used with small sample sizes (28). For all analyses, a negative ES represents that DEH impaired cognitive performance versus control conditions, whereas a positive ES represents an improvement.

The studies in the meta-analysis assessed a wide array of cognitive domains, providing several dependent outcomes (i.e., multiple tests with accuracy and reaction time) available to extract as results. Multiple ES are problematic for most conventional meta-analyses, as the dependent structure of results (e.g., decreased reaction time but increased accuracy) may confound and compromise validity of the results unless the covariance structure is known (29). Because of this, a multivariate (mixed-effects) meta-analysis was used. Multivariate meta-analyses are appropriate when multiple related outcomes are reported within each study (e.g., both reaction time and accuracy for a given test or multiple tests of executive function) and the dependence structure is unknown (30). Multivariate meta-analysis, compared with other techniques, can control for multiple outcomes without necessitating studywide averaging which can yield ES estimates that do not represent the range of study outcomes (30).

The meta-analysis was completed using the *rma.mv* function from the *metafor* package in R (www.metafor-project.org). The appropriate random effect structure was identified by fitting an intercept-only model (no moderators) with multiple random effect configurations. Using the *anova* function within R, each different random effect configuration was compared. The best random effect structure was identified from the model yielding the lowest Akaike information criterion. This process resulted in a random effects model which allowed modeled between-study differences along with within-study differences based upon cognitive domains and outcome variables (accuracy, reaction time) assessed.

To assess the overall effects of DEH on cognitive performance, an intercept-only model was used. Subgroup analysis was completed using the *mods* option within the *rma.mv* function. A Q test was instituted to examine if moderator variables significantly impacted the ES estimates. If the subgroup was categorical (e.g., ≤ or >2% BML), the ES estimates were compared with each other. If moderator variable was continuous (% BML), the slope was compared to zero using metaregression. Publication bias of studies included within the meta-analyses was assessed using a Duval and Tweedie trim and fill correction funnel plot from the *trimfill* function within *metafor*. Because the *trimfill* function cannot analyze multivariate meta-analysis structures, a random effects meta-analysis model was used for this analysis. Across all comparisons, an alpha level of ≤0.05 was used to indicate statistical significance. As is common practice, ES (Hedges g) of 0.2, 0.5, and 0.8 were considered small, moderate, and large, respectively, whereas ES < 0.1 considered trivial (31).

Subgroup analyses. We aimed to determine the influence of experimental factors on the overall ES using moderator analysis. A subgroup meta-analysis (i.e., meta-analyses comparing subsets of studies) was used to probe potential moderator variables such as the type of cognitive domain, type of performance outcome, method of DEH, or magnitude of DEH. To be considered a subgroup, we set a minimum of five studies required in the category. Study design was examined by classifying either pre–post (e.g., measurements compared from baseline to after intervention) or crossover (e.g., matched trials on separate days). Most studies (m = 23) assessed multiple domains of cognitive performance. The cognitive domains compared were attention, executive function, information processing, memory, motor coordination, and reaction time-specific tasks. Outcome variable types (accuracy and reaction time) across all cognitive tasks were also compared since most tasks provided both accuracy and reaction time outcomes.

Methods to induce DEH were coded into the following categories: exercise, heat exposure (ambient temperature, ≥27°C), exercise-heat stress (exercise + heat exposure with ambient temperature, ≥27°C), or fluid restriction. Two studies (32,33) induced DEH via both an exercise only and exercise plus diuretic trial. The administration of a diuretic did not significantly increase BM loss; therefore, both trials were averaged for the analysis. Because of the environment, both studies (32,33) were subsequently categorized as exercise-heat stress protocols. Another two studies used fluid restriction plus exercise to induce DEH (27,34). One study used a 15-h fluid restriction protocol followed by 45 min cycling at approximately 70% maximum effort in a temperate environment and was therefore classified as an exercise protocol (34). The other study (27) had subjects undergo a prolonged fluid restriction protocol before one measured condition followed by an exercise bout. In that case, the data point after exercise was classified as an exercise protocol.

The DEH methods were also categorized into two classifications: with/without the addition of environmental heat stress and with/without exercise. The magnitude of DEH was also subgrouped by cut point of ≤2% or >2% BML to examine whether cognitive studies inducing sufficient body water losses typically observed to elicit physiological compensation (5,35) had greater impairments (and parallel studies on physical performance). If information was provided about subject fitness level (based on either aerobic exercise testing or author description), this information was used to categorize

Copyright © 2018 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

subjects as sedentary, recreational, or highly fit ($\dot{V}O_2$ max > 55 mL·kg^{-1}·min^{-1}). In the presence of a significant Q value, pairwise comparisons were made between different levels of the moderator variable with Bonferroni–Holm corrections.

Metaregression. The magnitude of DEH (values ranging from 1.1% to 6.0% BML across individual studies) associated with cognitive task impairment was examined using metaregression. Because each specific BML was coded, multiple levels of DEH per study were possible, even with small differences (e.g., 2.1% vs 2.2%). Instances where raw BM measures were reported, a percent change score was calculated (%BML = (BMpost − BMpre)/BMpost). Aspects of environmental heat exposure (core temperature, duration of exposure > 27°C) were also analyzed using metaregression.

RESULTS

Study Characteristics

Supplementary Table 1 (see Table, Supplemental Digital Content 1, Characteristics of 33 studies, http://links.lww.com/MSS/B396) presents the characteristics of each study in the analysis. The final sample consisted of 33 studies (m), all were published in peer-reviewed journals except one (36) found in ProQuest. In total, there were 413 subjects and 280 ES (k) with a median of 6 ES per study (range: 1–36). All 33 studies used a repeated-measures (within subjects) design with 11 and 22 using a pre–post or crossover design, respectively. Practice trials or subject familiarization with the cognitive tasks were reported for 24 studies (9 did not specify). Across all studies, the median BML incurred was 2.1% (min–max, 1.1%–6.2%). Because DEH magnitude was determined based on %BML, nine studies had multiple levels within the study compared to 24 studies eliciting only one level of DEH. Sixteen studies elicited DEH via exercise-heat stress, seven using exercise only (no heat), three with heat stress only (no exercise), and four with fluid restriction only. Three studies used multiple methods of DEH.

The cognitive domains assessed after DEH were attention (m = 10; visual vigilance, test of variables of attention, monotonous driving task, oddball), executive function (m = 17; mental math, trail making test, proof reading, grammatical reasoning, map recognition, logical relation test), memory (m = 18; digit span, match to sample, n-back test, repeated acquisition, story recall, word recognition, map recall, picture recall), reaction time specific (m = 16; simple/choice reaction time), information processing (m = 8; perceptive discrimination, target evaluation, critical flicker fusion test, substitution test, visual perception test, letter-digit substitution), and motor coordination (m = 5; unstable tracking, manual tracking test, psychomotor test, Groton maze chase). Approximately half of the studies (17 of 33) reported at least one significant cognitive outcome impairment with DEH.

Of the 33 total studies, 27 (81%) included only male subjects. When fitness level was measured/described, only recreationally (m = 13) or highly fit (m = 13) subjects were included (7 studies did not specify).

Overall Effect of DEH on Cognitive Performance

Figure 2 presents the overall averaged ES for all cognitive performance outcomes within each of the 33 studies reporting effects of DEH. Considerable variation was observed among studies, with individual study-averaged ES ranging from −1.25 to 0.75. Nine (27%) studies demonstrated a study-averaged positive ES or improvement in cognitive performance (one was significant), whereas 23 (73%) studies had negative ES (8 were significant). When including all studies and outcomes (m = 33, k = 280), DEH elicited a small but significant impairment in cognitive performance (g = −0.21; P < 0.0001; 95% CI: −0.31 to −0.11). There was significant heterogeneity across studies (Q (279) = 696.0; P < 0.0001). The amount of total variance attributed to the total amount of within-study heterogeneity was low to moderate (I^2 = 37.6%), whereas between-study heterogeneity was low (τ^2 = 0.12). Study designs which used pre–post measures (resting control) within the same trial (g = −0.11, 95% CI: −0.29 to −0.07; P = 0.25) were not significant for cognitive impairments, although not significantly different (P = 0.15) compared with studies using crossover designs (g = −0.26; 95% CI: −0.38 to −0.14) which did show cognitive impairments overall (P < 0.001).

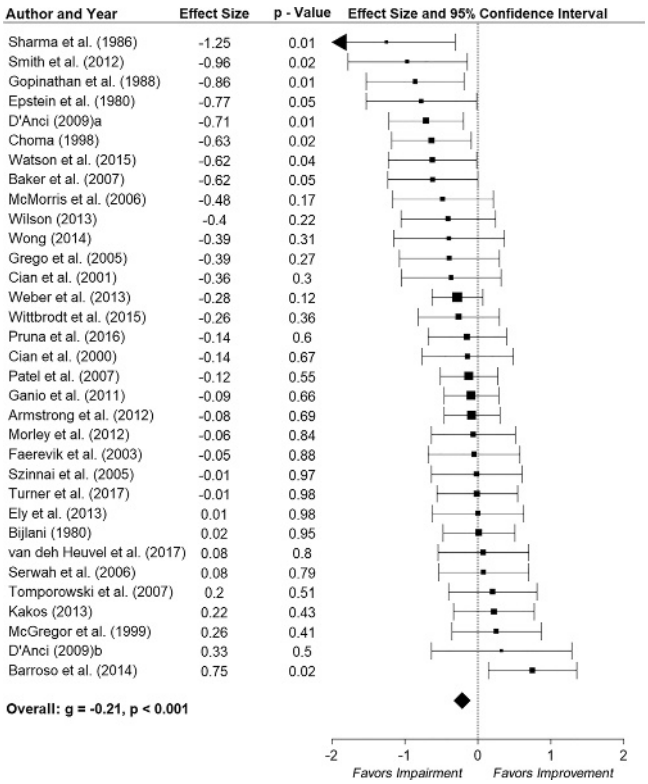

FIGURE 2—Forest plot of the effect size (ES) for all studies (m = 33) examining dehydration (DEH) on cognitive performance. Negative ES (g) indicate DEH impaired cognitive performance whereas positive ES (g) indicates improved cognitive performance. *Box size* indicates the relative weight of each study attributed to overall ES and horizontal lines indicate 95% confidence intervals. *Diamond* indicates mean overall ES with width corresponding to the 95% confidence interval.

APPLIED SCIENCES

Copyright © 2018 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

The trim and fill analysis suggested two additional DEH studies observing a strong positive effect (g of ~0.6, 0.9) on cognitive performance would be needed to minimize publication bias. However, the theoretical addition of these two positive ES studies did not alter significance of the overall ES (g = −0.17, 95% CI: [−0.29 to −0.05], P = 0.006), although the ES was reduced to between small and trivial. It is also highly unlikely that such a "theoretical" positive finding in improved cognitive function due to DEH would have remained unpublished. A subgroup meta-analysis comparing published studies versus unpublished studies was not possible since only one study was unpublished.

Analysis of Moderator Variables

Cognitive domains. Table 1 presents moderator variables for studies examining the effects of DEH on cognitive performance. DEH elicited a significantly greater impairment ($Q(1)$ = 5.6, P = 0.02) in accuracy compared to reaction time outcomes across the range of cognitive tests in all studies. There was a significant effect ($Q(5)$ = 51.9, P < 0.0001) of DEH when compared across the different broad categories of the cognitive domains assessed. Tasks assessing attention, executive function, and motor coordination had significant ($P \leq 0.01$) DEH-induced impairments but ES was not significant for information processing (P = 0.36), memory (P = 0.11), or reaction time-specific tasks (P = 0.14). The subgroup analysis indicated a greater impairment in tasks of attention (g = −0.54; 95% CI: −0.69 to −0.39; Q (1) = 31.5; P = 0.001) and motor coordination (g = −0.40; 95% CI: −0.63 to −0.17; $Q(1)$ = 14.6, P = 0.01) compared with reaction time specific tasks (g = −0.10; 95% CI: −0.23 to 0.03). Figure 3 presents the Forest plots for the subgroup analysis between tasks requiring attention (m = 10, k = 37), motor coordination (m = 5, k = 14), and those specific tasks based on reaction time (m = 16, k = 50). No other significant differences were found (P > 0.05) among the other categories (i.e., executive function, information processing, memory, and reaction time).

Level of DEH. Figure 4 presents the metaregression results for all ES estimates across the range of DEH. Overall, metaregression revealed a significant association between the magnitude of DEH (% BML) (slope = 0.07, $Q(1)$ = 4.0, P = 0.04) and decrements in cognitive performance, although the relationship (R^2 = 0.003) explained virtually none of the variance. Based on the subgroup analysis across all cognitive domains and outcomes, studies eliciting a BML >2% elicited significantly greater ($Q(1)$ = 4.2, P = 0.04) cognitive impairment than studies eliciting 2% BML, although ES estimates for both subgroups were significant (Table 1).

Other factors. No significant differences in ES were observed between the subgroups for methods to induce DEH ($Q(3)$ = 0.7, P = 0.87). Three methods (exercise, exercise-heat stress, and fluid restriction) elicited significant (P < 0.05) cognitive impairment while heat stress alone did not (P = 0.15). Subgroup analysis (Table 1) indicated that cognitive performance was not impaired to a greater extent (P = 0.54) when DEH was elicited with an element of environmental heat stress (heat stress or exercise-heat stress) compared with protocols without heat exposure (exercise only, fluid restriction).

TABLE 1. Effect of categorical moderators for the impact of DEH on cognitive performance.

Variables	*Q* Test, *P* Value	*g*	95% CI	*P*
Outcome variable	**0.02**			
Accuracy (m = 31, k = 185)		−0.25	[−0.35 to −0.14]	**<0.001**
Reaction time (m = 25, k = 95)		−0.14	[−0.23 to −0.05]	**0.006**
DEH magnitude	**0.04**			
≤2% BML (m = 18, k = 135)		−0.14	[−0.27 to 0.00]	**0.04**
>2% BML (m = 20, k = 145)		−0.28	[−0.41 to −0.16]	**<0.001**
Cognitive domains	**<0.0001**			
Attention (m = 10, k = 37)		−0.52*	[−0.66 to −0.37]	**<0.001**
Executive function (m = 17, k = 60)		−0.24	[−0.37 to −0.12]	**0.01**
Information processing (m = 8, k = 28)		−0.08	[−0.27 to 0.10]	0.36
Memory (m = 18, k = 91)		−0.10	[−0.22 to 0.02]	0.11
Motor coordination (m = 5, k = 14)		−0.40*	[−0.63 to −0.17]	**<0.001**
Reaction time tasks (m = 16, k = 50)		−0.10	[−0.23 to 0.02]	0.14
Method of DEH	0.87			
Exercise (m = 11, k = 43)		−0.21	[−0.38 to −0.04]	**0.01**
Heat stress, no exercise (m = 5, k = 31)		−0.17	[−0.39 to 0.06]	0.15
Exercise + heat stress (m = 16, k = 175)		−0.20	[−0.34 to −0.06]	**0.005**
Fluid restriction (m = 5, k = 33)		−0.29	[−0.53 to −0.06]	**0.02**
Exposure to heat (>27°C)	0.60			
Heat exposure (m = 21, k = 206)		−0.20	[−0.32 to −0.08]	**0.001**
No heat exposure (m = 15, k = 74)		−0.24	[−0.39 to −0.09]	**0.001**
Exposure to exercise	0.83			
Exercise presence (m = 24, k = 209)		−0.21	[−0.33 to −0.10]	**0.0002**
No exercise presence (m = 12, k = 71)		−0.20	[−0.36 to −0.03]	**0.02**
Subject fitness	0.31			
Recreationally fit (m = 13, k = 128)		−0.29	[−0.44 to −0.14]	**0.0001**
Highly fit (m = 13, k = 86)		−0.17	[−0.34 to 0.00]	**0.05**
Not specified (m = 7, k = 66)		−0.10	[−0.32 to 0.11]	0.36

Q Test *P* value: difference within a category, *g*: effect size (ES), 95% confidence interval (CI) of the ES, *P* value for the individual moderator compared to zero (*P*), *m*: number of studies, *k*: number of outcomes.
*Significantly (P < 0.01) different ES within cognitive domain compared to reaction time ES. DEH = dehydration.

Copyright © 2018 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

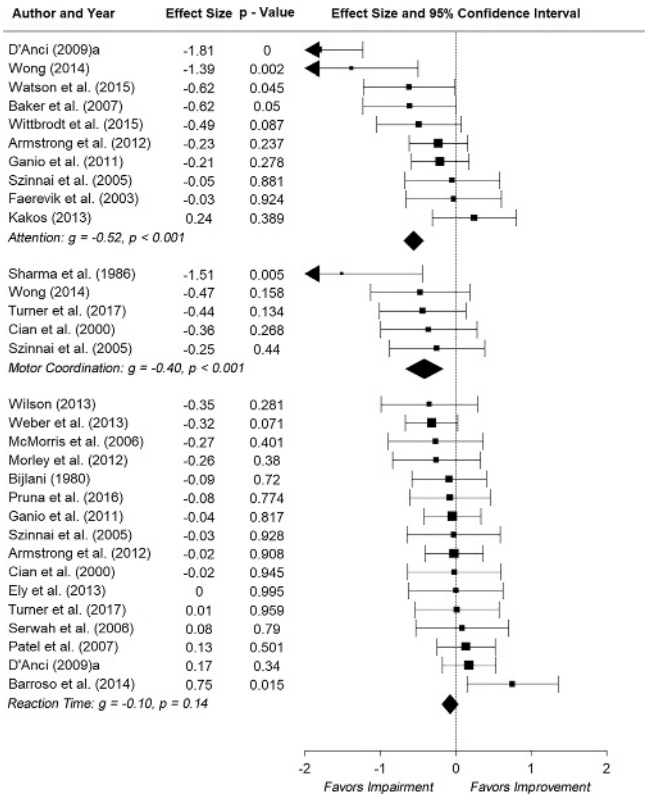

FIGURE 3—Forest plot of studies examining subgroup analysis for the effect of dehydration on tasks using different cognitive domains: attention (*top*), motor coordination (*middle*), and reaction time specific tasks (e.g., simple/choice reaction time, *bottom*). Effect size (ES) for attention ($Q(1) = 31.5$, $P < 0.001$) and motor coordination ($Q(1) = 14.6$, $P = 0.01$) were significantly greater than reaction time specific tasks. Negative ES (*g*) indicate DEH impaired cognitive performance whereas positive ES (*g*) indicates improved cognitive performance. *Box size* indicates the relative weight of each study attributed to overall ES and horizontal lines indicate 95% confidence intervals. *Diamond* indicates mean overall ES with width corresponding to the 95% confidence interval.

Furthermore, metaregression analysis did not yield significant associations between core temperature ($P = 0.78$, $R^2 = 0.005$) or duration of heat exposure ($P = 0.70$, $R^2 = 0.001$) to cognitive performance in those studies with heat exposure. Likewise, subgroup analysis comparing DEH methods using exercise to induce DEH (exercise, exercise-heat stress) were not different ($Q(1) = 0.05$, $P = 0.83$) in cognitive impairments compared to methods without exercise (Table 1). The subject level of fitness ($Q(2) = 2.3$, $P = 0.31$) did not differ among groups with both recreationally fit ($P = 0.0001$) and highly fit ($P = 0.05$) subjects experiencing significant cognitive impairment after DEH (Table 1).

DISCUSSION

Dehydration is believed to impair cognitive performance and potentially increase workplace accidents and occupational risk (37). Although narrative reviews suggest that DEH may impair cognitive performance (11,14), previous research has not uniformly supported this position (5). The current study used a quantitative analysis of studies and objectively determined that there is a significant effect of DEH on cognitive performance. We also assessed the influence of several study design factors that contribute to this effect. Our meta-analysis supports previous hypotheses (15,38) that some cognitive domains (i.e., attention, executive function, motor coordination) are more likely to degrade with DEH, especially when compared with lower-level tasks (i.e., reaction time) and also that the degree of cognitive impairment is associated with the magnitude of DEH.

The main finding of this study is that DEH elicits a small but significant impairment on cognitive performance. This negative ES aligns with narrative reviews, suggesting that DEH may mirror the effects of other nutritional interventions by altering cognitive performance but only by a small degree (13). Furthermore, this significant finding occurred in the face of significant study heterogeneity, which has been repeatedly acknowledged in narrative reviews (13,14), thus making a firm conclusion challenging. To this point, only 52% of studies (17 of 33) observed at least one statistically significant cognitive impairment after DEH. A meta-analytic technique overcomes this limitation and enhances the ability to assess the impact of various experimental factors potentially contributing to the heterogeneity of results.

The second main finding of the current study was that DEH does not affect all cognitive domains equally. Previous studies have demonstrated this experimentally (32,34,39) by observing significant cognitive impairments after DEH in some, but not all, cognitive domains. Some have also suggested higher-order cognitive domains may be impaired to a greater extent after DEH (11,14,15,38), although a proposed mechanism has not been identified. To our knowledge, this is the first comprehensive analysis to systematically demonstrate which

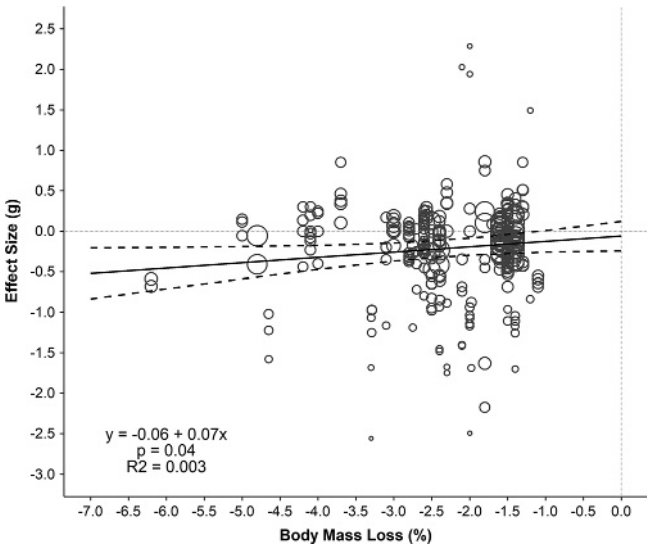

FIGURE 4—Metaregression analysis for the magnitude of body mass loss (%) and effect size (ES) (*g*) for cognitive task performance (negative values indicate impairment). Each outcome ($n = 280$) for all cognitive test variables is depicted by a circle, with the circle size representing the relative weight attributed to each ES. The slope for the line of best fit (*solid line*) was significantly different from zero ($P = 0.04$). *Dashed lines* indicated 95% confidence interval around line of best fit.

APPLIED SCIENCES

Copyright © 2018 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

cognitive domains may be more at risk after DEH. We observed that tasks requiring attention and/or executive function were significantly more impaired after DEH while others (e.g., information processing, memory, reaction time tasks) were not. Both attention and executive function are generally considered "higher-order" cognitive domains (40) and, as such, indicate more complex cognitive processing may be impaired following DEH. Previously, DEH (−1.6% BML) elevated frontoparietal brain activations during an executive function task (Tower of London) but without significant performance impairments (41), suggesting neural inefficiencies during task performance. Furthermore, because frontoparietal activations appear integral to executive functioning (40), prolonged cognitive processing (required for tests of attention), may also be responsible for executive function impairments. The specific rationale as to why other cognitive domains are less affected is yet to be understood. Specific cognitive domains may require different brain regions and neurotransmitter systems for adequate processing (11), potentially making some brain areas (and cognitive domains) more susceptible to body water deficits.

We also observed that tasks using motor coordination are significantly impaired following DEH to a greater extent than "lower-order" cognitive tasks. This is an analogous finding to multiple reports indicating skilled gross motor task degradation (e.g., basketball skills) following DEH (2,4). The tasks contained within the motor coordination domain in this meta-analysis largely consisted of fine motor (e.g., finger, hand) movements in response to visual stimuli sensed within the frontal eye field (42). Motor coordination tasks require neural processing in similar brain areas to attention/executive function tasks (e.g., frontal lobe), but also elicit activations within the motor thalamus, cerebellum, and basal ganglia (43). Why motor coordination may be more significantly impaired by DEH is not entirely clear. Because DEH appears to degrade perceptual responses of sensory mechanisms (e.g., thirst, hostility) (44), it is possible that the thalamus and basal ganglia, which also monitor these sensory systems, are uniquely challenged.

Along with differences in the effect of DEH on cognitive domains, we found that accuracy is impaired more than reaction time outcomes. This may suggest a change in strategy to preserve performance. One study demonstrated this experimentally after prolonged cycling, observing increased errors with DEH compared with faster reaction time (45) often referred to as the speed–accuracy tradeoff (46). An alternate explanation is that DEH simply impairs higher-order cognitive processes involved in decision making but, for reasons currently unclear, still elicit responses with similar temporal characteristics. Future studies might investigate how DEH alters speed–accuracy cognitive strategies.

Another major finding of this study was that cognitive performance declined along with the magnitude of water deficit and specifically, based on the subgroup analysis, when above 2% BML. This finding is in agreement with individual studies observing this graded phenomenon (6,7), but differs from others that have elicited large BML (~4% BML) without cognitive performance impairments (9,10). Furthermore, greater cognitive impairments were observed in studies eliciting a DEH threshold sufficient to induce physiological compensation (>2%) versus when compensation was unlikely (≤2% BML) (5,35). Taken together, these findings suggest that the hypovolemia (and subsequent physiological compensation) elicited by DEH may at least be partially responsible for cognitive impairments, and this effect is observed at increasingly greater BML. However, the level of body water deficit alone explained virtually none of the variance (<1%) in the ES estimates; thus, the mechanisms responsible for this effect are not entirely clear but likely relate to the multiple mitigating factors already cited (DEH method, cognitive domain, outcome measure) when a composite ES is used. It is well accepted that human cognitive capacity (i.e., ability to accomplish cognitive processing) is limited (47), and, as such, cumulative task demands which exceed a threshold level may result in performance decline. It is possible that progressive body water deficits (with increasing hypovolemia and thirst sensory distraction) may incrementally limit cognitive capacity resulting from a variety of mechanisms including altered neurotransmitter levels (11,48), or brain structures (41) known to be associated with degraded cognitive performance in aging and/or disease states (49).

Our analysis also attempted to rule out other factors influencing the impact of DEH on cognitive performance. Because increased fitness may increase cerebral circulation and brain perfusion (50), it was believed highly fit subjects may be more resilient to cognitive decrements following DEH. However, sedentary individuals were not recruited in these studies, or compared to recreationally or highly fit subjects. Future studies might investigate this factor with sedentary versus highly fit individuals using a nonexercise DEH protocol. The method used to achieve DEH also did not differentially influence the ES, although passive heating (without exercise) did not elicit significant cognitive impairments. This finding conflicts with narrative reports, suggesting heat stress may be required to elicit cognitive deficits (15). Some have suggested that, when DEH is coupled with exercise and/or heat stress, the "true" effect of DEH is confounded (11). Effect size estimates for all DEH methods were small to moderate (~−0.2 to −0.5), suggesting any obfuscation of the "true" effect of DEH on cognitive performance by multiple physiological stressors is likely minor. Furthermore, three of the 33 studies (27,39,51) have investigated multiple DEH methods within a single-study testing the same subjects and concluded similar results. Two studies reported cognitive impairments with both exercise and heat stress alone (39,51) and another found no difference between fluid restriction or fluid restriction combined with exercise (27).

As has been suggested previously, meta-analyses have some inherent limitations (28). A meta-analysis does not allow for mechanistic explanations but can provide a framework to guide future investigations. Although not directly assessed in this study, DEH-mediated cognitive impairments may be influenced by affective changes, such as altered mood

Copyright © 2018 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

(32,33) or the presence/sensation of thirst (44,52) contributing to the total allocation of neural resources. Mental exertion may also be elevated during cognitive testing (26,41,53); however, this has not always paralleled performance impairments. These perceptual measures have been reviewed previously (13), but merit future meta-analytic investigation. Another inherent limitation is that some studies (8,39,54,55) omitted reporting data from nonsignificant tests after DEH. It is also possible that not all studies were of similar quality in terms of randomization, double blinding, and convenience sampling. Thus, the impact of these limiting factors on the current meta-analysis is unclear, and we acknowledge that conclusions may change as future studies appear in the literature.

In conclusion, we have identified that, despite many studies using different experimental protocols reporting a variable range of results, DEH elicits a small, but significant impairment in cognitive performance. Furthermore, high-order cognitive processing (involving attention and executive function) and motor coordination appear more susceptible to impairment following DEH compared with other domains involving lower-order mental processing (e.g., simple reaction time). The magnitude of DEH is associated with the impairment in cognitive performance, specifically notable when >2% BML. Thus, the threshold for the impact of DEH on cognitive performance may be similar to that previously reported for the performance of physical exertional tasks (i.e., exercise).

The authors would like the acknowledge the contributions of Morgan Arnold, Michael Jones, Hayley Keadey, Rachel McAllister, and Lauren Pitz during the systematic review and data extraction.

All authors had no conflict of interest, including relevant financial interests, activities, relationships, and affiliations to declare relating to this manuscript. The results of the study are presented clearly, honestly, without fabrication, falsification, or inappropriate data manipulation. The results of the present study do not constitute endorsement by the American College of Sports Medicine.

REFERENCES

1. Sawka MN, Burke LM, Eichner ER, Maughan RJ, Montain SJ, Stachenfeld NS. American College of Sports Medicine position stand. Exercise and fluid replacement. *Med Sci Sports Exerc*. 2007;39:377–90.
2. Lindseth PD, Lindseth GN, Petros TV, Jensen WC, Caspers J. Effects of hydration on cognitive function of pilots. *Mil Med*. 2013;178(7):792–8.
3. Savoie FA, Kenefick RW, Ely BR, Cheuvront SN, Goulet ED. Effect of hypohydration on muscle endurance, strength, anaerobic power and capacity and vertical jumping ability: a meta-analysis. *Sports Med*. 2015;45(8):1207–27.
4. Baker LB, Dougherty KA, Chow M, Kenney WL. Progressive dehydration causes a progressive decline in basketball skill performance. *Med Sci Sports Exerc*. 2007;39(7):1114–23.
5. Cheuvront S, Kenefick R. Dehydration: physiology, assessment, and performance effects. *Compr Physiol*. 2014;4:257–85.
6. Gopinathan PM, Pichan G, Sharma VM. Role of dehydration in heat stress-induced variations in mental performance. *Arch Environ Health*. 1988;43:15–7.
7. Sharma VM, Sridharan K, Pichan G, Panwar MR. Influence of heat-stress induced dehydration on mental functions. *Ergonomics*. 1986;29(6):791–9.
8. Adam G, Carter R, Cheuvront S, et al. Hydration effects on cognitive performance during military tasks in temperate and cold environments. *Physiol Behav*. 2008;93:748–56.
9. Ely B, Sollanek K, Cheuvront S, Lieberman H, Kenefick R. Hypohydration and acute thermal stress affect mood state but not cognition or dynamic postural balance. *Eur J Appl Physiol*. 2013;113:1027–34.
10. van den Heuvel AMJ, Haberley BJ, Hoyle DJR, Taylor NAS, Croft RJ. The independent influences of heat strain and dehydration upon cognition. *Eur J Appl Physiol*. 2017;117:1025–37.
11. Lieberman H. Hydration and cognition: a critical review and recommendations for future research. *J Am Coll Nutr*. 2007;26: 555S–61S.
12. Lieberman HR. Methods for assessing the effects of dehydration on cognitive function. *Nutr Rev*. 2012;70(2 Suppl):S143–6.
13. Masento NA, Golightly M, Field DT, Butler LT, van Reekum CM. Effects of hydration status on cognitive performance and mood. *Br J Nutr*. 2014;111:1841–52.
14. Grandjean A, Grandjean N. Dehydration and cognitive performance. *J Am Coll Nutr*. 2007;26(5 Suppl):549S–54S.
15. Nuccio RP, Barnes KA, Carter JM, Baker LB. Fluid balance in team sport athletes and the effect of hypohydration on cognitive, technical, and physical performance. *Sports Med*. 2017;47(10):1951–82.
16. King J. Brief account of the sufferings from a detachment of United States Cavalry, from deprivation of water, during a period of eighty-six hours, while scouting on the "Llano Estacado" or "Staked Plains." *Texas Am J Med Sci*. 1878;75:404–68.
17. Adolph E. *Physiology of Man in the Desert*. New York: Interscience; 1947. pp. 137–41.
18. Lieberman H, Bathalon G, Falco C, Kramer M, Morgan C, Niro P. Severe decrements in cognition function and mood induced by sleep loss, heat, dehydration, and undernutrition during simulated combat. *Biol Psychiatry*. 2005;57(4):422–9.
19. Krause AJ, Simon EB, Mander BA, et al. The sleep-deprived human brain. *Nat Rev Neurosci*. 2017;18(7):404–18.
20. Strachan MW, Deary IJ, Ewing FM, Ferguson SS, Young MJ, Frier BM. Acute hypoglycemia impairs the functioning of the central but not peripheral nervous system. *Physiol Behav*. 2001;72(1–2):83–92.
21. Opstad PK, Ekanger R, Nummestad M, Raabe N. Performance, mood, and clinical symptoms in men exposed to prolonged, severe physical work and sleep deprivation. *Aviat Space Environ Med*. 1978;49(9):1065–73.
22. Benton D, Young H. Do small differences in hydration status affect mood and mental performance? *Nutr Rev*. 2015;73(2 Suppl):83–96.
23. Lezak MD, Howieson DB, Loring DW, Fischer JS. *Neuropsychological Assessment*. New York: Oxford University Press; 2004. p. 1039.
24. Strauss E, Sherman EMS, Spreen O. *A Compendium of Neuropsychological Tests: Administration, Norms, and Commentary*. New York: Oxford University Press; 2006. p. 1235.
25. Wittbrodt MT, Millard-Stafford M, Sherman RA, Cheatham CC. Fluid replacement attenuates physiological strain resulting from mild hypohydration without impacting cognitive performance. *Int J Sport Nutr Exerc Metab*. 2015;25:439–47.
26. Watson P, Whale A, Mears SA, Reyner LA, Maughan RJ. Mild hypohydration increases the frequency of driver errors during a prolonged, monotonous driving task. *Physiol Behav*. 2015;147:313–8.
27. Weber AF, Mihalik JP, Register-Mihalik JK, Mays S, Prentice WE, Guskiewicz KM. Dehydration and performance on clinical concussion measures in collegiate wrestlers. *J Athl Train*. 2013;48(2):153–60.
28. Borenstein M, Hedges LV, Higgins JPT, Rothstein HR. *Introduction to Meta-Analysis*. 1st ed. Chichester, UK: Wiley; 2009. p. 452.

Copyright © 2018 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.

29. Scammacca N, Roberts G, Stuebing KK. Meta-analysis with complex research designs: dealing with dependence from multiple measures and multiple group comparisons. *Rev Educ Res*. 2014;84(3):328–64.
30. Becker BJ. Multivariate Meta-Analysis. In: Tinsley EAH, Brown SD, editors. *Handbook of Applied Multivariate Statistics and Mathematical Modeling*. London: Academic Press; 2000. pp. 499–525.
31. Durlak JA. How to select, calculate, and interpret effect sizes. *J Pediatr Psychol*. 2009;34(9):917–28.
32. Ganio MS, Armstrong LE, Casa DJ, et al. Mild dehydration impairs cognitive performance and mood of men. *Br J Nutr*. 2011;106:1535–43.
33. Armstrong LE, Ganio MS, Casa DJ, et al. Mild dehydration affects mood in healthy young women. *J Nutr*. 2012;142:382–8.
34. Patel AV, Mihalik JP, Notebaert AJ, Guskiewicz KM, Prentice WE. Neuropsychological performance, postural stability, and symptoms after dehydration. *J Athl Train*. 2007;42(1):66–75.
35. Cheuvront SN, Ely BR, Kenefick RW, Sawka MN. Biological variation and diagnostic accuracy of dehydration assessment markers. *Am J Clin Nutr*. 2010;92:565–73.
36. Kakos LS. Improving cognitive function following exercise-induced dehydration: role of sports drink supplementation [dissertation]. Kent (OH): Kent State University; 2013. p. 124.
37. Kenefick R, Sawka M. Hydration at the work site. *J Am Coll Nutr*. 2007;26(5 Suppl):597S–603S.
38. Tomporowski PD, Beasman K, Ganio MS, Cureton K. Effects of dehydration and fluid ingestion on cognition. *Int J Sports Med*. 2007;28:891–6.
39. Cian C, Barraud PA, Melin B, Raphel C. Effects of fluid ingestion on cognitive function after heat stress or exercise-induced dehydration. *Int J Psychophysiol*. 2001;42(3):243–51.
40. Logue SF, Gould TJ. The neural and genetic basis of executive function: attention, cognitive flexibility, and response inhibition. *Pharmacol Biochem Behav*. 2014;123:45–54.
41. Kempton M, Ettinger U, Foster R, et al. Dehydration affects brain structure and function in healthy adolescents. *Hum Brain Mapp*. 2011;32(1):71–9.
42. Murray EA, Bussey TJ, Wise SP. Role of prefrontal cortex in a network for arbitrary visuomotor mapping. *Exp Brain Res*. 2000;133(1):114–29.
43. Bosch-Bouju C, Hyland B, Parr-Brownlie L. Motor thalamus integration of cortical, cerebellar and basal ganglia information: implications for normal and parkinsonian conditions. *Front Comput Neurosci*. 2013;7:163.
44. Engell DB, Maller O, Sawka MN, Francesconi RN, Drolet L, Young AJ. Thirst and fluid intake following graded hypohydration levels in humans. *Physiol Behav*. 1987;40:229–36.
45. Grego F, Vallier JM, Collardeau M, Rousseu C, Cremieux J, Brisswalter J. Influence of exercise duration and hydration status on cognitive function during prolonged cycling exercise. *Int J Sports Med*. 2005;26:27–33.
46. Fitts PM. The information capacity of the human motor system in controlling the amplitude of movement. *J Exp Psychol*. 1954;47(6):381–91.
47. Wickens CD. Multiple resources and performance prediction. *Theor Issues Ergon Sci*. 2002;3(2):159–77.
48. Cheuvront SN, Carter R, Kolka MA, Lieberman HR, Kellogg MD, Sawka MN. Branched-chain amino acid supplementation and human performance when hypohydrated in the heat. *J Appl Physiol*. 2004;97(4):1275–82.
49. Persson J, Nyberg L, Lind J, et al. Structure–function correlates of cognitive decline in aging. *Cereb Cortex*. 2006;16(7):907–15.
50. Davenport MH, Hogan DB, Eskes GA, Longman RS, Poulin MJ. Cerebrovascular reserve: the link between fitness and cognitive function? *Exerc Sport Sci Rev*. 2012;40(3):153–8.
51. Cian C, Koulmann N, Barraud PA, Raphel C, Jimenez C, Melin B. Influence of variations in body hydration on cognitive function: effect of hyperhydration, heat stress, and exercise-induced dehydration. *J Psychophysiol*. 2000;14:29–36.
52. Egan G, Silk T, Zamarripa F, et al. Neural correlates of the emergence of consciousness of thirst. *Proc Natl Acad Sci U S A*. 2003;100:15241–6.
53. Szinnai G, Schachinger H, Arnaud MJ, Linder L, Keller U. Effect of water deprivation on cognitive-motor performance in healthy men and women. *Am J Physiol Regul Integr Comp Physiol*. 2005;289:R275–80.
54. Wong SH, Sun FH, Huang WY, Chen YJ. Effects of beverages with variable nutrients on rehydration and cognitive function. *Int J Sports Med*. 2014;35(14):1208–15.
55. Choma CW, Sforzo GA, Keller BA. Impact of rapid weight loss on cognitive function in collegiate wrestlers. *Med Sci Sports Exerc*. 1998;30:746–9.
56. Baker LB, Conroy DE, Kenney WL. Dehydration impairs vigilance-related attention in male basketball players. *Med Sci Sports Exerc*. 2007;39:976–83.
57. Barroso S da S, Almeida RD de, Gonzaga W da S, et al. Hydration status and cognitive-motor performance during a fast triathlon race in the heat. *Rev Educ Física UEM*. 2014;25(4):639–50.
58. Bijlani R, Sharma K. Effect of dehydration and a few regimes of rehydration on human performance. *Indian J Physiol Pharmacol*. 1980;24(4):255–66.
59. D'Anci KE, Vibhakar A, Kanter JH, Mahoney CR, Taylor HA. Voluntary dehydration and cognitive performance in trained college athletes. *Percept Mot Skills*. 2009;109(1):251–69.
60. Epstein Y, Keren G, Moisseiev J, Gasko O, Yachin S. Psychomotor deterioration during exposure to heat. *Aviat Space Environ Med*. 1980;51:607–10.
61. Faerevik H, Reinertsen RE. Effects of wearing aircrew protective clothing on physiological and cognitive responses under various ambient conditions. *Ergonomics*. 2003;46(8):780–99.
62. McGregor SJ, Nicholas CW, Lakomy HKA, Williams C. The influence of intermittent high-intensity shuttle running and fluid ingestion on the performance of a soccer skill. *J Sports Sci*. 1999; 17(11):895–903.
63. McMorris T, Swain J, Smith M, et al. Heat stress, plasma concentrations of adrenaline, noradrenaline, 5-hydroxytryptamine and cortisol, mood state and cognitive performance. *Int J Psychophysiol*. 2006;61(2):204–15.
64. Morley J, Beauchamp G, Suyama J, et al. Cognitive function following treadmill exercise in thermal protective clothing. *Eur J Appl Physiol*. 2012;112:1733–40.
65. Pruna GJ, Hoffman JR, McCormack WP, et al. Effect of acute L-alanyl-L-glutamine and electrolyte ingestion on cognitive function and reaction time following endurance exercise. *Eur J Sport Sci*. 2016;16(1):72–9.
66. Serwah N, Marino FE. The combined effects of hydration and exercise heat stress on choice reaction time. *J Sci Med Sport*. 2006;9:157–64.
67. Smith MF, Newell AJ, Baker MR. Effect of acute mild dehydration on cognitive-motor performance in golf. *J Strength Cond Res*. 2012;26:3075–80.
68. Turner JM, Marsteller DA, Luxkaranayagam AT, Fletcher JM, Stachenfeld NS. Mild exercise in female subjects impairs complex learning independent of hydration status and emotion. *Physiol Behav*. 2017;180:113–119.
69. Wilson G, Hawken MB, Poole I, et al. Rapid weight-loss impairs simulated riding performance and strength in jockeys: implications for making-weight. *J Sports Sci*. 2014;32(4):383–91.

Copyright © 2018 by the American College of Sports Medicine. Unauthorized reproduction of this article is prohibited.